IonCure

A PRACTICAL PREPAREDNESS GUIDE

DISASTER-READY DOON

LANDSLIDES, EARTHQUAKES & FOREST FIRES

An evidence-led guide for families, schools,
hotels and neighbourhoods in Dehradun, Uttarakhand

Research conducted with the support of

IonCure Tech Pt. Ltd.

A practical, evidence-led preparedness resource

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First edition • October 2026

DEHRADUN • UTTARAKHAND

Disaster-Ready Doon: Landslides, Earthquakes & Forest Fires

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Important: read before you rely on this book

This book is general educational guidance. It is not an official emergency instruction, an engineering assessment, a legal opinion or medical advice. In an emergency, follow instructions from the Uttarakhand State Disaster Management Authority, the Dehradun district administration, the police, fire services, the Forest Department and other authorised responders. Where this book and an official instruction differ, follow the official instruction.

Emergency numbers, standards, alert systems and risk classifications change. Every such item is date-stamped and must be checked against the current official source before you act on it or reprint it.

No map, drawing or diagram in this book is an engineering or geospatial risk assessment. Hazard at a specific site can only be judged by qualified professionals using current data.

No expert named here has reviewed or endorsed this book.

Illustrations are original and schematic. Quoted figures are attributed to their sources in the Source Register at the back; items marked VERIFY are not yet confirmed against a primary document.

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Before you begin

The thesis. Most preparedness advice fails in a predictable place: the moment it meets an ordinary night. The power is out, the phone is at nine per cent, the school bus is on the wrong side of a washed-out culvert, and a forwarded message claims an earthquake is due at eleven. A plan that depends on any one of those things working is not a plan. It is a hope with a laminated cover.

This book therefore asks a narrower question than most guides do: *what will a real household, classroom or hotel front desk in Dehradun actually do, with what they can afford, understand and maintain, when the usual systems fail?* Science comes first because bad science produces bad habits. But every chapter ends in something you can do, check or print.

The promise. You will leave with an honest picture of three hazards, a method for testing your own building, route and routine, a set of checklists that work on paper, and a 30/90-day plan. You will also leave with a clear sense of what nobody can promise: no one can predict an earthquake, and no map can tell you that your particular slope is safe.

> **How evidence is labelled in this book**
>
> **Established** – supported by peer-reviewed work or an official source, cited in the chapter.
>
> **Reported** – stated in news or a press release; useful, but not independently verified.
>
> **Proposal** – the author's idea, tested here only on paper. Feasibility, cost and authority requirements are stated; none is an official recommendation.
>
> **VERIFY** – a fact that must be confirmed against a primary source before publication or use.

What this book is not. It is not a substitute for a structural engineer, a geologist, the Forest Department or the emergency services. Wherever one of them is required, the text says so plainly. It also does not claim that every part of the Doon valley faces the same hazard. A historical hazard location is not a dangerous one today without current evidence, and a regional study is not a verdict on your street.

How to read it. Start with Chapters 1 and 5. Then go to the hazard that worries you most, and to the checklists in the Appendices.

CHAPTER 1

Understanding the Doon landscape and its risks

Reader outcome

You can name the three hazards, say when each is most likely, and explain the difference between a hazard and a risk.

Dehradun sits in a valley between the Shivalik foothills and the Lesser Himalaya, with the Mussoorie range rising on its northern edge. That sentence is standard geography, and it is also the whole problem in miniature: a city built where mountains are still being made. The rocks along the Dehradun–Mussoorie highway are cut by the Main Boundary Thrust and by many folds and faults, as geologist S. C. Roy described in a study of landslide hazard along the road alignments (S9). The Himalayan Frontal Thrust, the fault system at the mountain front, has been studied for evidence of past great earthquakes (S19).

None of this means disaster is imminent. It means the ground, the slopes and the forests of the Doon follow rules that residents rarely see, and that those rules operate on three different clocks.

1.1 Hazard is not risk

A *hazard* is the thing that can happen: ground shaking, a debris flow, a fire. *Risk* is what the hazard does to people and property, and it depends on three further questions. Who and what is exposed? How vulnerable are they: an unreinforced wall, a drain choked with plastic, a school with one exit? And what capacity exists to cope, from a neighbour with a ladder to a district control room that answers its phone? Two houses on the same street can face the same shaking and very different risk.

This is why the book keeps returning to buildings, drains, routes and habits. Hazard is largely outside our control. Exposure, vulnerability and capacity are not.

1.2 Three hazards, three clocks

Earthquakes have no season. Forest fire has one, and the Forest Department treats 15 February to 15 June as the fire season (S13). Landslides and floods follow the monsoon, which in this region broadly runs from mid-June to September (approximate; check the current IMD onset and withdrawal dates). Most households prepare for one clock and forget the others.

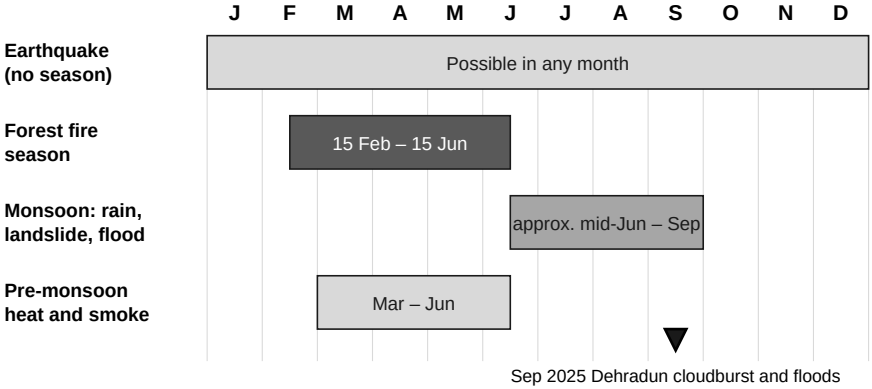


Figure 1.1 The annual hazard calendar for the Dehradun area. Fire season dates: Uttarakhand Forest Department (S13). Monsoon span is approximate and illustrative. Original diagram; not an official hazard calendar.

1.3 What the classification says, and does not say

Under the seismic zoning in IS 1893 (Part 1):2016, Dehradun is listed in Zone IV, the second-highest of four design zones (S1, S3). A revised map notified in November 2025 added a new Zone VI for the Himalayan arc, but the Bureau of Indian Standards withdrew that revision by gazette notification on 3 March 2026 and reinstated the 2016 standard, according to press reports (S2). **VERIFY** against the BIS gazette before printing, because this is exactly the kind of fact that changes.

A zone is a design instruction to engineers. It is not a forecast, and it does not say when or whether a damaging earthquake will occur.

1.4 Case study: 15–16 September 2025

What happened. Heavy rain and a reported cloudburst struck Dehradun district overnight on 15–16 September 2025. The district disaster management office named Sahastradhara, Maldevta, Santla Devi and Dalanwala as worst hit, and recorded 192 mm of rain at Sahastradhara and 141.5 mm at Maldevta (S5). Roads were breached and bridges washed away; Mussoorie and Sahastradhara were cut off by landslides (S5, S6).

What the sources establish. The state authority reported 13 deaths in Dehradun district and 16 people missing on 16 September (S5). A later report put the statewide toll at 21 (S5b). Hundreds of students trapped by waterlogging at an institute campus in Paundha were rescued by SDRF; reports give 200 and 400–500 for the number (S5, S6).

What remains uncertain. Final casualty counts, the exact rainfall at other locations, and the role of riverbank construction. Local anger over riverbank construction was reported in October 2025 (S21); the causal claim has not been verified here.

What readers can reasonably learn. Rain was local and intense; roads, not only buildings, failed; and a single campus of children was stranded by water that had nowhere to go. The lessons are about routes, drainage, and release procedures, and they recur in Chapters 3 and 7.

Sources: S1, S2, S3, S5, S6, S9, S13, S19, S21.

CHAPTER 2

Earthquake readiness and safer buildings

Reader outcome

You can explain why buildings, not shaking, cause most casualties, run a basic walk-through of your own home, and know what to do during and after an earthquake.

2.1 What the ground is doing

The Indian plate is pushing under the Tibetan plateau along the Main Himalayan Thrust, and strain builds up between earthquakes (S19). Great Himalayan earthquakes such as Kangra (1905, Mw 7.8) are part of this record. Prof. Javed Malik and colleagues at IIT Kanpur study how past earthquakes ruptured the frontal thrust, using trenches and dating; their work on the Kumaon foothills reports paleoseismic evidence of events in CE 1505 (a date the authors mark as uncertain) and CE 1803 (S19). In July 2025 he told the press that building infrastructure directly on the frontal thrust "can be harmful" and that an active-fault atlas was due (S19).

Two cautions. First, paleoseismology reconstructs the past; it does not schedule the future. Second, work from the Kumaon segment cannot be transferred to Dehradun without local evidence. What the Doon reader can safely take away is plain: strong shaking is a *when-prepared* question, not a *whether-prepared* one.

2.2 What makes a building safer

Shaking does not kill people. Falling buildings, parapets, water tanks and cabinets do. Engineers designing to IS 1893 rely on ductility, the ability of a structure to bend and dissipate energy without sudden collapse. For reinforced concrete frames, IS 13920 prescribes the detailing that provides it, and it applies in Zones III to V (S3; **VERIFY** clause).

A non-engineer cannot certify a building, but anyone can notice the features that make engineers worry. Use the walk-through below to prepare questions for a qualified structural engineer. It is a conversation starter, not an assessment.

Household walk-through: ten things to notice

1. An open ground floor used for parking under living floors (a soft storey).
2. Floors or rooms added later, especially without drawings or an engineer.
3. Heavy water tanks, solar heaters or planters on a thin roof edge.
4. Tall unbraced parapets, chimneys, boundary walls and gate pillars.
5. Diagonal or stair-step cracks in walls, or cracks that have grown.
6. Brick or block infill with gaps, or walls that rattle when you push.
7. Beds and cribs under heavy shelves, mirrors or hanging fixtures.
8. A slope, retaining wall or river edge within a few metres of the house.
9. Blocked stairs, grilles with no key, or a single exit.
10. Gas cylinders unsecured, or no way to shut off gas and power quickly.

2.3 Old houses and Koti Banal: what to take, what to leave

The village of Koti Banal in the Yamuna valley gave its name to a distinct building tradition of timber-reinforced stone masonry found in Rajgarhi area, Uttarkashi district. Investigators report that the region evolved this style as early as about 1,000 years ago and that it reached its peak around 880 years ago, that it uses locally available wood, stone and slate, and that its features resemble modern earthquake-resistant design, with performance reported in earthquakes such as 1905 Kangra and 1991 Uttarkashi (S11, S12). The researchers also note that surviving structures are deteriorating from neglect and surrounding development (S11, S12).

This book uses Koti Banal as a *Himalayan comparison*. It is not evidence that old houses in Dehradun are generally safe. Doon's older buildings were made in different conditions, from different materials, and many have been altered.

Question	Answer
Tested enough to deserve engineering validation?	Raised platform, simple rectangular plans, timber bands tying walls, separation from neighbours.
Cannot be safely transferred?	Copying the look on a modern brick or concrete extension; adding floors to an old wall; treating age as proof.
Folklore or unproven?	"Old houses never fall." The record shows some did not; it does not show all did not.
Who decides?	A structural engineer with experience of masonry and timber.

2.4 During and after the shaking

Widely taught protective action is **Drop, Cover and Hold On**: get low, take cover under a sturdy table or against an interior wall away from glass, and hold on until the shaking stops (general guidance; cross-check the current USDMA and NDMA wording before print). Do not run out while the ground is moving, and do not use lifts. Outdoors, move away from buildings, walls, poles, trees and slopes.

Afterwards, expect aftershocks. Check yourself and others for injury, check for gas smell, and switch off gas and electricity only if you can do it safely. Leave a damaged building and do not re-enter until it has been inspected. Use text messages rather than calls to keep lines free for emergencies.

Sources: S1, S2, S3, S11, S12, S19.

CHAPTER 3

Landslides and the Dehradun–Mussoorie corridor

Reader outcome

You can describe why the corridor is landslide-prone, read a susceptibility map without over-reading it, and decide what to do when rain and warning signs combine.

3.1 Why the corridor slides

The Dehradun–Mussoorie area combines Krol and Tal formations of the Mussoorie Syncline, with the Main Boundary Thrust and many folds and faults affecting the rock along the highway (S9). Roy recorded many landslides, particularly near the Kalagad and Nalota nalas, and warned that mining and mining traffic add instability (S9). More recent work links landslides in Mussoorie to rapid urbanisation and climate change (S8).

Recent evidence is more specific. A rockslide on the Dehradun–Mussoorie highway on 9 July 2022 left a scarp of nearly 80 m above road level, and debris extended 70 m below it (S10). Such slopes are not new, but the people and vehicles beneath them are.

3.2 Reading a susceptibility map honestly

The Wadia Institute of Himalayan Geology mapped landslide susceptibility across 84 km² of Mussoorie and its surroundings. About 15 per cent of the area was rated highly susceptible, 29 per cent moderate and 56 per cent low to very low; the very high and high zones lay mainly in settled areas (S7). A 2024 study of 18 road-cut slopes found four unstable, three near Jharipani and one near Hathi Paon Road, while slopes around the Junu waterfall were stable (S8).

Three rules for using a susceptibility map

It describes terrain, not events. “High susceptibility” means conditions favour sliding if triggered. It does not mean a slide is due.

It is a regional tool. It cannot clear or condemn your plot; an engineering-geological survey can.

It ages. New road cuts, drains and construction change a slope after the map was drawn.

The same caution applies to place names. Naming a location as historically affected does not make it dangerous today, and its absence from a list does not make it safe.

3.3 Warning signs, rain and a decision

Commonly listed field signs of slope trouble include new cracks in ground or walls, tilting trees or poles, bulging or leaning retaining walls, sudden muddy or changed spring and stream water, and unusual rumbling (general landslide guidance; cross-check NDMA material before print). Heavy rain on an already wet slope is the common trigger, and the September 2025 event shows how quickly it can turn.

PROPOSAL: a household rain-and-slope diary

Idea. Keep a dated notebook of heavy-rain days, new cracks, blocked drains and water-colour changes, with phone photographs where possible.

Feasibility and cost. Free; five minutes a week. **Limits.** It cannot forecast a slide and must never delay leaving when signs appear. **Authority.** None needed to keep a diary; show it to municipal or geology officials as a record, not as evidence of risk.

Status. Untested in Dehradun.

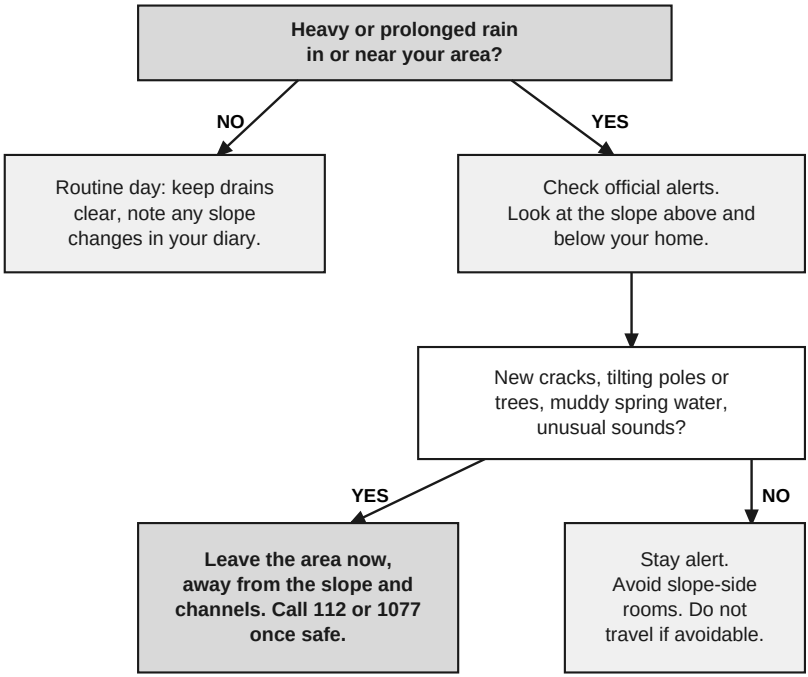


Figure 3.1 A decision path for rain and slope signs. Original diagram; a prompt for thinking, not an official protocol.

3.4 On the road

For commuters and tourists the corridor is the exposure. Check the road status and the forecast before leaving. In heavy rain do not stop beneath cut slopes, near drainage channels or on bridges. Never drive or walk into flowing water. If the road is blocked, turn around when you can do so safely and do not try to cross a fresh slide.

Alternatives such as the Kimadi or Kimiari (Lambhidhar) and Kiarkuli routes exist, but Roy found landsliding near the Main Boundary Thrust on both and argued for sharing traffic across the alignments (S9). An alternative route is a different risk, not an absence of risk.

Sources: S5, S6, S7, S8, S9, S10.

CHAPTER 4

Forest fires, Rajaji, Thano and settlement-level preparedness

Reader outcome

You can explain what drives the Himalayan fire season, why satellite alerts mislead, what a household near a forest edge can safely do, and where the evidence is still missing.

4.1 The season in numbers

The Forest Department treats 15 February to 15 June as the fire season (S13). By 26 May 2026 the state forest minister reported 394 fire incidents covering about 331 hectares in 2026, and 14,638 incidents covering 23,682.77 hectares over the previous ten years (S13). Season totals swing widely. Department data reported in May 2026 showed 162 incidents in April alone, against 184 in April 2025 and 724 in April 2024 (S13). The 2026 hotspot was the Badrinath–Kedarnath belt, not Dehradun (S13). Quick Response Teams were stationed at seven sensitive locations, including Dehradun and Mussoorie (S13).

The usual picture is not a crown fire racing through old forest. Reporting and research describe mostly human-ignited surface fires between roughly 300 and 2,000 m, fed by dry leaves and chir pine needles (S14). A regression of 19 years of MODIS data found that pre-monsoon temperature, the share of chir pine, and lower pre-monsoon and winter rainfall together explained 56 per cent of the variation in fire density between forest divisions (S14). A CSIR-IHBT satellite study likewise linked chir-pine forests to nearly 57 per cent of incidents across the Western Himalaya (S15). A Forest Research Institute study has also mapped fire-risk zones for the Raipur range of Mussoorie Forest Division (S22).

4.2 Why alerts mislead

Satellite alerts are a gift and a trap. Between 1 November 2025 and 20 January 2026 the Forest Department received 1,952 alerts; field checks confirmed only 132 as actual fires, about 6 to 7 per cent, affecting 18.84 hectares (S13). The department raised the matter with the Forest Survey of India. The lesson for households is not to distrust all alerts. It is that an alert is a cue to check an official source, not a confirmation of fire near you, and that a flood of unverified screenshots helps nobody.

4.3 What a household near a forest edge can safely do

The aim is to reduce what can ignite near the building, and to be ready to leave, never to fight a fire. Everything below is ordinary housekeeping that needs no authorisation on your own property. Anything on forest land, including fire lines, back-burning or cutting trees, belongs to the Forest Department.

- Clear dry leaves, pine needles and litter from roofs, gutters, balconies and the yard before mid-February, and again during the season.
- Store firewood, cylinders and fuel away from the walls and from vegetation.
- Never burn waste, crop stubble or leaves during the fire season, and never discard cigarettes or fireworks on dry ground.
- Keep water, a hose or buckets, and a torch where you can reach them; keep the way out clear.
- Know who to call: report a fire to the fire service or the Forest Department, and use the state and district numbers in Appendix C.
- Leave when instructed. Take people, pets, medicines and documents. Do not return until cleared.

> **Do not**
>
> Do not light a “counter-fire” or clear forest litter by burning. Do not enter a burning forest to help. Do not try to extinguish anything larger than a small, contained fire at your own property. Community fire lines and vegetation management must follow authorised Forest Department guidance.

4.4 Smoke, schools and vulnerable people

Smoke travels farther than flame. Children, older adults, pregnant women and people with asthma, heart or lung disease are most affected. Staying indoors with windows closed helps. N95-type respirators can reduce fine-particle exposure when well fitted; a wet handkerchief does not filter well (general public-health guidance; **VERIFY** wording). People with respiratory conditions should ask their doctor in advance for a smoke plan. Schools should treat heavy-smoke days like heat days and adjust outdoor activity.

4.5 Rajaji and Thano: an evidence gap, stated plainly

The brief for this book names the Rajaji National Park and the Thano landscape. The research completed for this edition did not locate a verified, current dataset on fire frequency, ignition sources, or settlement exposure specifically for Rajaji and the Thano range. So this book does not make claims about them. What can be said is general: forest edges, villages and roads at the boundary of protected areas are places where people and fuel meet, and where wildlife may be displaced by fire.

> **Open questions for the Forest Department and park authorities**
>
> What are the recorded fire counts and areas for the Dehradun-side Rajaji and Thano ranges, by year?
>
> Which villages sit within reach of the forest edge, and what is their evacuation route?
>
> How should residents report wildlife encountered during a fire?
>
> How can residents take part in authorised prevention and early reporting?

Sources: S13, S14, S15, S22.

CHAPTER 5

Early warnings, official information and misinformation

Reader outcome

You can tell monitoring, early warning, forecast and prediction apart, know what warnings exist in Uttarakhand, and check a message before you act or forward it.

5.1 Four words that are not synonyms

Term	Meaning	Time scale	Useful for
Monitoring	Instruments detect events and conditions as they happen.	Continuous	Knowing an earthquake occurred; rain, river and slope data.
Early warning	A fast alert after an event begins, sent ahead of damaging waves.	Seconds to a minute	Drop-cover-hold, stopping lifts and trains, shutting valves.
Forecast	A probability of conditions over a period, such as heavy rain or fire weather.	Hours to days	Closing roads, pre-positioning crews, deciding to travel.
Prediction	A specific time, place and size of a future event.	Not available	Nothing: no earthquake can currently be predicted this way.

Anyone who offers you an earthquake date is offering a prediction, and a prediction is not currently possible. Treat every such message as false.

5.2 What exists in Uttarakhand

IIT Roorkee, sponsored by the Uttarakhand State Disaster Management Authority, built the Uttarakhand state earthquake early warning system (UEEWS), described in 2023 as comprising about 170 seismic sensors with a central server at Roorkee (S16). An app, “Uttarakhand Bhookamp Alert”, was launched by USDMA on 4 August 2021. It issues warnings for damaging earthquakes originating in Uttarakhand, and it depends on an internet connection (S16). Low-cost sirens were installed in two major cities and in district emergency operations centres (S16). A later IIT Roorkee app, BhuDEV, has been reported (S16; **VERIFY** its current status).

How much warning? A researcher told the press that Dehradun might get about 11 seconds for a Himalayan earthquake, against about a minute in Delhi, and that this is “not enough to completely evacuate buildings” (S17, reported, undated). Eleven seconds is enough to drop, cover and hold. It is not enough to leave a building.

5.3 Weather, rain and fire information

Heavy-rain warnings come from the India Meteorological Department, and during the September 2025 event schools in Dehradun were closed under a red alert (S6b, reported). The Forest Department has said it signed an agreement with IMD for fire-weather forecasting (S13). **VERIFY** which official channels are current before listing them in a print edition.

When power and mobile networks fail, information is still possible: battery or hand-crank radio, community radio where available, loudspeaker announcements by authorities, and neighbours with working phones. A plan that depends only on an app has already failed.

5.4 A three-question check for any message

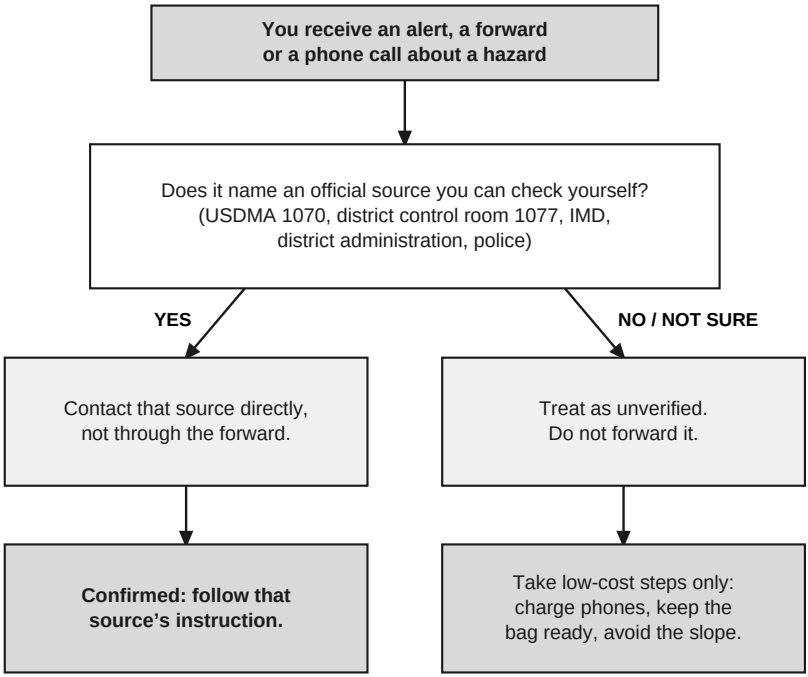


Figure 5.1 Warning-information flowchart. Original diagram.

Ask three questions in order. *Who says so?* Name the person or office. *Can I check it myself?* Call or look it up, rather than replying to the forward. *What does it cost to be wrong?* Moving away from a slope on a false alarm costs an evening; ignoring a true one can cost a life. Low-regret actions are acceptable before verification. Forwarding is not.

PROPOSAL: the paper-first card

Idea. One laminated A5 card per household with the verified numbers, the two meeting points, and the three-question check, kept by the door.

Feasibility and cost. A few rupees to print; no phone needed. **Maintenance.** Re-check every February and every June. **Limits.** A card cannot update itself; date-stamp it.

Status. Untested; a printable template is in Appendix B.

Sources: S6b, S13, S16, S17.

CHAPTER 6

Family preparedness, evacuation and the disaster kit

Reader outcome

You can build a family plan that survives a power cut and a dead phone, assemble a kit, and include everyone in it.

6.1 What fails first

Most plans quietly assume electricity, mobile coverage, internet, roads, water and accurate official information. Test the plan by removing them one at a time.

If this fails	What breaks	Low-cost workaround
Electricity	Lights, lifts, pumps, phone charging, routers.	Torches and spare batteries; charged power banks; know the stairs.
Mobile network	Calls, apps, maps, online alerts.	Pre-agreed meeting points; texts before calls; paper card; battery radio.
Internet	Early-warning app, messaging groups, maps.	Printed contact list and local route sketch.
Roads	School pick-up, ambulances, supplies.	Pre-decide who collects children; keep a stock for several days.
Water supply	Drinking, cooking, hygiene.	Stored drinking water; know the nearest safe source.
Official information	Clarity on what to do.	Two trusted sources; the three-question check (Chapter 5).

6.2 A plan on one page

- Two meeting places: one just outside the home and one outside the neighbourhood.
- One out-of-area contact everyone can text.
- Who collects which child, and what happens if that person cannot reach the school.
- The safest spot to take cover in each room, and a quick review of heavy objects that can fall.
- How to shut off gas and electricity, and who is physically able to do it.

- Which two routes you would use to leave, and which one you would not use after heavy rain.
- What to do with pets, medicines, documents and cash.

Evacuation planning questions are worth asking before the day: *Who will tell us to go? Where is the assigned shelter? Can we reach it on foot? How will we carry what we need? Who in the building cannot manage stairs?* Do not leave because of rumour, and do leave when authorities instruct you to.

6.3 The household kit

A study in Uttarakhand found that although national agencies recommend preparedness kits, their usefulness for survivors had not been empirically tested in the region, and that the state action plan treats kits as a response measure rather than a preparedness one (S18). The kit below is therefore a reasoned minimum, not an official standard.

Item	Why	Maintain
Drinking water in sealed containers	Supply may stop; hydration comes first.	Rotate every six months.
Non-perishable food, opener	Shops and roads may be unavailable.	Check dates each season.
Torch, spare batteries, power bank	Dark, no charging.	Test monthly; fresh batteries yearly.
Battery or hand-crank radio	Information without a network.	Test monthly.
First-aid kit	Cuts, burns, sprains while help is delayed.	Replace expired items.
Regular medicines (one week), prescriptions	Gaps can be dangerous.	Rotate with refills.
Photocopies of IDs, insurance, key numbers	Phones may fail.	Update after any change.
Whistle, sturdy shoes, mask (N95 type)	Signal; protect feet and lungs from debris and smoke.	Keep by the bed.
Cash in small notes	Digital payments may stop.	Check yearly.
Warm clothing, blanket, rain cover	Hill nights are cold.	Swap by season.

6.4 Everyone in the plan

Make the plan work for the people most likely to be left out. **Infants and young children:** formula or milk supplies, nappies, comfort items, and a trusted adult for each child. **Older adults:** medicines, spare spectacles or hearing-aid batteries, and a named neighbour who will check on them. **People with disabilities:** backup mobility aids, ways to receive alerts that are not only audible or only visual, and a person who knows how to assist. **Pets:** food, water, leash or carrier, and vaccination records.

The buddy list is old technology and still the best. Write down who will check on whom, and tell them. A person who agrees in advance is easier to find than a person you hope to find.

Sources: S18.

CHAPTER 7

Schools, hotels and institutions

Reader outcome

You can design and test a drill, state a hotel's responsibilities, and prepare a school or hotel kit that fits its real constraints.

7.1 Schools: the stranded-campus problem

During the September 2025 floods, about 200 to 500 students were trapped by waterlogging at the Devbhoomi Institute campus in Paundha and were rescued by the SDRF (S5, S6; reports differ on the number). The sources do not say whether the campus had a flood plan. They do show the failure mode: a large group, a blocked road, and water in the wrong place.

A school plan therefore needs more than a fire drill. It needs separate procedures for earthquake, fire, flood or road closure, and heavy smoke. It needs a rule for when children are released and when they are kept, a parent communication method that works without the internet, and named adults for children who need help. A 2024 study recommended reviving the National School Safety Programme and helping children build preparedness kits (S18).

7.2 Hotels and tourism operators

Hotels are responsible for people who do not know the area, may not speak the local language and arrive without a kit. In the September 2025 floods tourists were staying in hotels and no deaths among them were initially reported (S6, reported); that is a statement about one morning, not a standard.

- A written evacuation plan and a floor plan with exits posted in each room, in the languages guests use.
- A guest register that can be taken outside quickly, with a head-count routine.
- Staff who know the shut-offs for gas, power and water and can use them safely.
- Structural advice from a qualified engineer before adding floors, balconies or water tanks.
- A rule for road closures: when to stop guests travelling and where to accommodate them.
- A first-aid kit and at least one trained staff member on every shift.

7.3 Kits for institutions

The tables in Appendix A give separate checklists for a household, a school and a hotel. They explain what to store, why it matters, and how to maintain it. They are practical minimums, not official standards.

7.4 Designing a drill that teaches

A drill that has never failed has not tested anything. Run an unannounced scenario with a twist: the main exit is blocked, the headteacher is absent, the phones are off. Observe with a checklist, time each stage, and write down three things to change. Include children with disabilities and visitors. Report the results to staff within a week, and repeat the drill after the changes.

> **PROPOSAL: the blackout drill**
>
> **Idea.** Once a term, run a 30-minute drill in which phones are collected in a box, lights are off in the corridor and one exit is declared blocked.
>
> **Feasibility and cost.** No cost beyond time; needs head teacher approval and parental notice. **Risks.** Panic and falls; keep it in daylight, brief children first, assign marshals. **Status.** Untested here; adapt with the school's safety officer and local fire service.

Sources: S5, S6, S18.

CHAPTER 8

Neighbourhood resilience and authorised community prevention

Reader outcome

You can map what your neighbourhood has and lacks, know what residents may and may not do about drains, slopes and fire, and start a low-cost community record.

8.1 A neighbourhood asset and vulnerability map

Before any disaster, a lane or colony knows more than any agency about itself: who is elderly and alone, whose house is leaning, which tank holds water, who is a nurse or an electrician. Write it down on paper. Share people's details only with consent and keep the list with one or two trusted coordinators.

- Mark two assembly points and the lane's exits on a hand-drawn sketch.
- List skills and tools: first aid, ladder, stretcher, vehicles, generator, radio.
- List neighbours who need help to leave, with their consent, and pair each with a buddy.
- Agree a phone tree and a paper notice board, in case the group chat goes silent.

8.2 Drains, slopes and rivers

Rain becomes a hazard when it has nowhere to go. Residents can keep their own drains and gutters free of waste, avoid dumping debris on slopes and channels, and report blocked or broken public drains and culverts to the municipal authority. They should not cut slopes, dig or fill near streams, or build or alter retaining walls without qualified advice and the required permissions. The 2025 reports of local anger over riverbank construction show how strongly residents feel about the issue (S21), although this book makes no claim about any particular site.

8.3 Fire: joining authorised efforts

Prevention begins with not starting fires and reporting them quickly. Anything more must follow Forest Department direction. A researcher at a Dehradun non-profit has argued in the press that too many rules keep people away from forests (S14b, reported), which is a policy debate rather than a licence. The safe course is to ask the local forest range office how residents can take part in reporting, awareness and authorised vegetation work, and to do only what is authorised and supervised.

8.4 Recording what you see

PROPOSAL: the neighbourhood hazard diary

Idea. A shared, dated log of observations: blocked drains, new cracks, rainfall events, smoke days, road closures, with photos.

Feasibility and cost. A notebook or shared spreadsheet; ten minutes a week for one volunteer. **Benefits.** Gives municipal and geological officials a record, and helps residents see patterns.

Limits. It is not a monitoring system, cannot predict events and must not be used to assure safety. Avoid recording personal data. **Status.** Untested.

Sources: S14b, S21.

CHAPTER 9

Response, recovery and a 30/90-day action plan

> **Reader outcome**
> You can act sensibly in the first hours, begin recovery safely, and follow a dated plan that turns this book into habits.

9.1 The first hours

Check for injuries and immediate danger, move to your meeting point, and follow official instructions. Help only in ways you are trained for. Do not enter damaged buildings, cross floodwater, approach fires or touch fallen power lines. Use texts rather than calls, preserve phone battery, and share verified information only.

9.2 Recovery

- Photograph damage before cleaning, and keep receipts; insurers and authorities may ask.
- Have a qualified engineer inspect any damaged building before it is re-occupied.
- Check on neighbours, especially anyone alone, injured, elderly or disabled.
- Expect disrupted sleep and strong emotions, especially in children. If distress persists, seek help from a doctor or a trained counsellor.
- Restock the kit and update the plan while the lessons are fresh.

9.3 Learning afterwards

Hold a short family or staff review within a month. What worked? What did we not know? Which contact was wrong? Write the changes on the paper card and update the date. A review that does not change a document, a habit or a stock of supplies has not happened.

9.4 The 30/90-day action plan

When	Do this	Done
Days 1–7	Write the paper card; verify emergency numbers with the official sources in Appendix C; choose two meeting points; do the household walk-through (Chapter 2).	[ ]
Days 8–30	Assemble the kit (Appendix A); clear gutters and yard litter; agree buddy arrangements; run one blackout drill at home or at school.	[ ]
Days 31–60	Ask a structural engineer about any concern from the walk-through; find the district disaster management plan; meet neighbours and draw the lane map.	[ ]
Days 61–90	Review drill results and change one thing; start the hazard diary; set calendar reminders for 15 February, 15 June and the monsoon onset.	[ ]

Preparedness is not a feeling of safety. It is a short list of things that are written down, stocked, tested and dated. Start with the card.

Appendix A: Kit checklists for household, school and hotel

Each table says what to store, why it matters and how to maintain it. Quantities depend on headcount; check current NDMA and USDMA guidance before print. These are practical minimums, not official standards.

Household (a family of four as a planning example)

Store	Why	Maintain
Water, food, openers, cooking fuel (safe storage)	Supply gaps	Rotate
Torches, batteries, power banks, radio	Dark, no network	Test monthly
First-aid kit, regular medicines, spectacles	Delayed help	Replace expired
IDs and insurance copies, cash	Phones may fail	Update yearly
Items for infants, older adults, disabled, pets	Different needs	Review each season

School

Store	Why	Maintain
Class lists, parent phone list on paper, release register	Accounting for every child	Update each term
Torches, whistles, megaphone, radio	Direction in noise and dark	Test monthly
First-aid kit and a stretcher or improvised alternative	Injuries before help arrives	Replace expired
Water and a few days of dry food (where feasible)	Stranding by flood or road closure	Rotate
Plan for children with disabilities and medical needs	Equal safety	Review each term
Smoke-day and flood-day procedures	Fire-season and monsoon	Review in Feb and June

Hotel

Store	Why	Maintain
Guest register and evacuation map in guest languages	Account for visitors	Update daily
Torches, radios, power banks, first-aid kits per floor	Guests lack kits	Check monthly
Water, blankets, food for stranded guests	Road closures	Rotate
Shut-off instructions and staff roles	Fast, safe response	Train each season
Contact list: ambulance, fire, police, USDMA, local clinic	Rapid help	Date-stamp, verify

Appendix B: Printable one-page card

Cut out, fill in and laminate. Re-check every February and every June.

Family emergency card	
Address and landmark	
Meeting point 1 (near home)	
Meeting point 2 (outside the area)	
Out-of-area contact	
Doctor and medicines	
Buddy for older adult or disabled person	
Three-question check	Who says so? Can I check it myself? What does it cost to be wrong?
Last checked (date)	

Appendix C: Emergency contacts (date-stamped)

Prepared 21 October 2026. Numbers 1070, 1077 and 0135-2710334 are shown on the USDMA website (S4); the page's own update date was not visible. Other numbers are widely published national or state numbers that must be verified before print.

Service	Number	Status
State Emergency Operation Centre (USDMA), toll free	1070	Shown on USDMA site, 21 Oct 2026
State Emergency Operation Centre (USDMA), landline	0135-2710334	Shown on USDMA site, 21 Oct 2026
District Emergency Operation Centre	1077	Shown on USDMA site, 21 Oct 2026
National emergency response	112	VERIFY
Ambulance	108	VERIFY
Fire	101	VERIFY
Police	100	VERIFY
Nearest hospital (fill in)		Local, to be added
School or ward contact (fill in)		Local, to be added

Appendix D: Living researchers to consult

The following people have recent, verifiable work relevant to this book. None has reviewed or endorsed it. An interview is a possibility, not an arrangement. Status shown is the evidence found on 21 October 2026; confirm current roles before contact.

Person	Relevance and evidence	Where to read
Piyoosh Rautela, disaster management researcher	Lead author of the Koti Banal report (S11); author of a case study on cloudbursts at Maldeota, Dehradun district (title listed, year VERIFY); wrote for Outlook in Oct 2024.	world-housing.net (report 150); dergipark.org.tr/en/pub/@sushil
Javed N. Malik, IIT Kanpur	Active tectonics and paleoseismology of the Himalayan Frontal Thrust; quoted in July 2025 press (S19).	iitk.ac.in/new/javed-n-malik
Kamal, IIT Roorkee	Developed the UEEWS earthquake early warning system with his team (S16, 2023).	news.careers360.com (2023 report)
Amit Kumar Verma, Forest Research Institute	Author of a Western Himalaya fire-driver study with IIT Roorkee, press report undated (S14b); VERIFY role.	thenewsminute.com article on Himalayan forest fires
Praveen Maghelal (with Sudha Arlikatti)	Heliyon 2024 study of household kits and flash-flood resilience in Uttarakhand (S18).	doi.org/10.1016/j.heliyon.2024.e41446

Appendix E: Glossary

Hazard / risk. What can happen / what it does to people and property.

Susceptibility. How favourable terrain is to a hazard, not a forecast.

Ductility. A structure's ability to deform without sudden collapse.

MBT / HFT. Main Boundary Thrust / Himalayan Frontal Thrust: major fault systems.

EEW. Earthquake early warning: seconds of notice after rupture begins.

SDRF / USDMA. State Disaster Response Force / Uttarakhand State Disaster Management Authority.

DEOC. District Emergency Operation Centre.

Appendix F: Source register

Verification: **P** = primary or official page read; **S** = secondary report; **V** = VERIFY before publication. Retrieved 21 Oct 2026.

ID	Source	Use	Status
S1	MoES, Lok Sabha Unstarred Q 563, 19 Jul 2017 (moes.gov.in)	Dehradun in Zone IV (2002 edition listing)	P
S2	Press: Vision IAS / ET summary; BharatNotes (Mar 2026)	IS 1893:2025 withdrawn 3 Mar 2026	S, V
S3	infralens.in guides to IS 1893:2016	Zone IV; IS 13920 applicability	S, V
S4	usdma.uk.gov.in site header	Contact numbers	P, V date
S5	Deccan Herald / PTI, 16 Sep 2025; Ada Derana, 17 Sep 2025	Sep 2025 event, rainfall, deaths	S
S5b	Deccan Herald, 17 Sep 2025	Statewide toll of 21 (reported)	S
S6	Tribune / PTI, 16 Sep 2025	Rescue and tourist details	S
S6b	India TV News, 16 Sep 2025	Schools closed under red alert	S
S7	DST release, 31 Aug 2020 (J Earth Syst Sci study)	WIHG Mussoorie susceptibility	P
S8	Mandal et al., Discover Geoscience 2024, doi 10.1007/s44288-024-00055-9	Mussoorie road-cut slopes	P
S9	S. C. Roy, J Nepal Geol Soc 28:95–100	Geology of the corridor; year VERIFY	P, V
S10	Chandel et al., EGU General Assembly 2023	9 Jul 2022 rockslide	P
S11	Rautela et al., World Housing Encyclopedia report 150	Koti Banal	P
S12	Rautela and Joshi, Disaster Prev Manage 2009; Current Science 2008	Koti Banal antiquity	P (abstracts)
S13	ETV Bharat 30 Jan, 6 May 2026; UNI 26 May 2026; Patrika 17 Apr 2026; Statesman	Fire season data	S
S14	Singh et al., Tropical Ecology 65(1):32–42 (2024)	Drivers of fires	P (abstract)
S14b	The Week, 24 Jun 2022; TheNewsMinute (undated)	Fire commentary	S, V
S15	CSIR-IHBT via PreventionWeb	Chir pine share	S
S16	Careers360 20 Mar 2023; OpenGov Asia; Rathore et al. preprint	UEEWS and app	S, V
S17	PreventionWeb / PTI (M. L. Sharma)	11-second claim	S, V

ID	Source	Use	Status
S18	Maghelal and Arlikatti, Heliyon 2024, doi 10.1016/j.heliyon.2024.e41446	Kits and schools	P
S19	IIT Kanpur profile; EGU24-1056; ETV Bharat 19 Jul 2025	HFT paleoseismology	P, S
S21	News Arena headline, 15 Oct 2025	Local concern on riverbank construction	S, V
S22	Advances in Forest Fire Research 2018 (FRI Dehradun)	Pine-zone fire risk, Mussoorie division	P